



# **PLANT PROTECTION ASSOCIATION OF INDIA**

Golden Jubilee Celebrations (1972 - 2022) & Continuing Legacy

PJTAU Campus, Rajendranagar, Hyderabad - 500030, India

## **OFFICIAL PROCEEDINGS REPORT**

# **ENHANCING AGRICULTURAL CONTRIBUTION AND GDP GROWTH**

## **A Comprehensive Policy, Regulatory, and Technological Roadmap for India**

Convened by:

**The Plant Protection Association of India (PPAI)**

**&**

**In Collaboration and Support by Dhanuka Agritech Limited**

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## **SESSION DETAILS**

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Date of Session: June 6, 2026  
Venue: The Taj Krishna, Banjara Hills, Hyderabad, Telangana  
Compiling Editors: **Dr. Sarath Babu Balijepalli, Dr. Gururaj Katti, Dr. T.V.K. Singh**  
Target Audience: Ministry of Agriculture & Farmers' Welfare, ICAR, SAUs, NAARM, and Industry Stakeholders

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# Enhancing Agricultural Contribution and GDP Growth - A Comprehensive Policy, Regulatory and Technological Roadmap for India

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## Abstract

*A structured short-, medium-, and long-term execution roadmap (0-20 years) was established to transition India from subsistence farming to innovation-driven, globally competitive, and sustainable agriculture. The institutional blueprints generated from these proceedings serve as an actionable guide for government bodies and industry stakeholders to protect farmer incomes, assure input quality, and bolster agricultural contributions to the national economic growth.*

**Keywords:** Plant Protection, Agricultural GDP, Policy Reform, Micro-Irrigation, Spurious Inputs, Pharma-Grade Auditing, Crop Diversification, Viksit Bharat @2047.

## Executive Summary

**(Integrated Proceedings of the Brainstorming Session on Enhancing Agricultural Contribution and GDP Growth)**

**Event Context:** Integrated Proceedings of the Brainstorming Session organized by the Plant Protection Association of India (PPAI) in collaboration with Dhanuka Agritech Ltd. (6 June 2026, Taj Krishna, Hyderabad).

**Background:** Indian agriculture contributes approximately 16-18% to the national GDP while supporting nearly half of the country's population. Despite its foundational role, the sector faces compounding bottlenecks, including soil degradation, irrigation mismanagement, climate vulnerability, intensifying pest pressures, low farmer awareness, and the proliferation of counterfeit agricultural inputs. To address these challenges and align with the national vision of Viksit Bharat @2047, the Plant Protection Association of India (PPAI), in collaboration with Dhanuka Agritech Ltd., convened a high-level brainstorming session featuring a distinguished panel of policymakers, agricultural scientists, vice-chancellors, and industry leaders.

## Methodology & Key Deliberations

The dialogue integrated macroeconomic perspectives with agronomic advancements to chart a structured, phased roadmap. Experts evaluated the following five core themes:

**Water & Irrigation Resilience:** Formulating a micro-irrigation (MI) strategy as a national water security pillar.

**Crop Diversification:** Diverting 10-15% of paddy areas to high-value pulses and oilseeds.

**Research & Innovation:** Advocating for public-private partnerships (PPP) and increasing national agricultural research investment to 1.5% of GDP.

**Market & Farmer Returns:** Establishing localized mandis and digitizing middleman registries.

**Regulation & Quality Control:** Curbing unregistered microbials and ensuring crop-group-wise Maximum Residue Limits (MRLs).

## **Major Outcomes and Policy Directions**

The session culminated in a two-layered institutional blueprint aimed at structural transformation:

**A Regulatory Framework for Crop Protection:** Introduction of a pharma-grade auditing model for crop-protection manufacturing plants. This includes Good Manufacturing Practices (GMP) standards, automated batch logging, unannounced audits, and stage-gate traceability powered by QR-coded batch fingerprinting and KYC-linked licensing to entirely eliminate spurious inputs.

**The Micro-Irrigation (MI) Strategy:** A 5-pillar deployment model prioritizing digital precision technologies (IoT, sensors) and extending MI infrastructure directly into canal-irrigated command areas to radically optimize water-use efficiency.

**Actionable Research Network:** Leveraging Krishi Vigyan Kendras (KVKs) and Farmer Producer Organizations (FPOs) to deploy community seed banks, scale digital diagnostics via a National Plant Protection Diagnostic Network, and launch a National Seed Treatment Mission.

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# Integrated Proceedings of the Brainstorming Session on Enhancing Agricultural Contribution and GDP Growth - A Comprehensive Policy, Regulatory and Technological Roadmap for India

The Plant Protection Association of India (PPAI), in collaboration with Dhanuka Agritech Ltd., convened an exclusive Brainstorming Session on Enhancing Agricultural Contribution and GDP Growth on 6th June 2026 at the Taj Krishna, Banjara Hills, Hyderabad.



Figure 1. Expert Group of Panelists

This high-level dialogue was organized by PPAI, blending **macroeconomic insights** with **agronomic advancements** to chart a comprehensive roadmap for strengthening India's agricultural sector. The session was designed to provide actionable recommendations for government, industry, and research institutions, with a focus on **policy reform, farmer returns, sustainability, and productivity**.

The event was graced by eminent experts and thought leaders, including:

- Dr. R.G. Agarwal, Chairman Emeritus, Dhanuka Agritech Ltd.
- Dr. C.D. Mayee, Former Chairman, ASRB & President, South Asia Biotechnology Centre
- Dr. P.K. Chakrabarthy, Former Member, ASRB
- Dr. B. Venkateswarlu, Former Director, CRIDA
- Dr. V. Praveen Rao, Vice-Chancellor - Kaveri University, Former VC, PJTAU
- Dr. K.S. Varaprasad, Former Director, ICAR-IIOR
- Dr. Y.G. Prasad, Former Director, ICAR-CICR
- Dr. R.K. Mathur, Director, ICAR-IIOR
- Dr. Gopal Lal, Joint Director, ICAR-NAARM
- Dr. Shaik N. Meera, Director, ICAR-ATARI, Zone X
- Dr. P. Partha Sarathy, Agricultural Economist



Figure 2. Expert Panel During Brainstorming

- Dr. K. Kareemulla, Former Principal Scientist, ICAR-NAARM
- Dr. T.V.K. Singh, Former Dean, PJTAU
- Dr. Gururaj Katti, Former Principal Scientist, ICAR-IIRR
- Dr. B. Parameswari, General Secretary, PPAI
- Dr. Bhaskar Bajar, Scientist (Sr. Scale), Executive Member, PPAI
- Dr. P. Spruthika Reddy, CEO, FPO
- Dr. Venkateswar Rao, Programme Coordinator, KVK Jammikunta
- Shri K. Ramesh Reddy, Senior GM, Dhanuka Agritech Ltd.
- Dr. M. Parusuramaiah, Senior DGM, Dhanuka Agritech Ltd.
- Dr. B. Sarath Babu, President Plant Protection Association of India

The collective expertise of this distinguished panel ensured that the deliberations were both **technically rigorous** and **policy-relevant**, laying the foundation for a unified roadmap to enhance India's agricultural GDP contribution and farmer prosperity.

## Opening

**Dr. B. Sarath Babu (President, PPAI)** welcomed participants and emphasized the Association's role in advancing plant protection and agricultural innovation. **Dr C.D. Mayee** took lead in organizing the day's brainstorm.

## Keynote Presentation - Dr. R.G. Agarwal

- Evolution of national slogans: "*Jai Jawan, Jai Kisan*", "*Jai Vigyan*", "*Jai Anusandhan*".
- Agriculture contributes **16-18% to GDP**, industry 24-27%, services 52-54%.
- Key challenges: soil degradation, policy fatigue, irrigation mismanagement, climate variability, pest pressure, counterfeit inputs, low farmer awareness.
- Solutions: stronger public-private partnerships, demand-driven crop planning, improved irrigation, market reforms, awareness campaigns.
- Vision for **Viksit Bharat @2047**: strengthen agriculture, promote innovation, empower farmers.



*Figure 3. Dr R G Agarwal Delivering Keynote Talk*

## Expert Recommendations

### Water & Irrigation

- Mandate micro-irrigation for horticulture (**Dr. C.D. Mayee**).
- Subsidies >50% for drip irrigation in water-stressed districts (**Dr. V. Praveen Rao**).
- Promote crop-specific irrigation packages for rice/wheat.
- Encourage artificial groundwater recharge structures (**Dr. B. Venkateswarlu**).

**A Comprehensive Strategy under this head was proposed by Dr. V. Praveen Rao:**

- Pillar 1: Recognize Micro-Irrigation (MI) as a National Water Security and Climate Resilience Strategy.
- Pillar 2: Prioritize MI deployment in Water-Stressed, Drought-Prone, and High-Value Crop Zones.
- Pillar 3: Reform and streamline Subsidy Norms for Small, Marginal, and Vulnerable Farmers to accelerate adoption.
- Pillar 4: Promote Smart and Digital Irrigation Technologies (IoT, sensors, precision delivery).
- Pillar 5: Extend MI infrastructure directly into Canal-Irrigated Command Areas to optimize water-use efficiency.

### Crop Diversification & Value Addition

- Incentives for dryland farmers to grow pulses/oilseeds (**Dr. K. Kareemulla**).
- Shift 10-15% of paddy area to pulses/oilseeds (**Dr. B. Venkateswarlu**).
- Promote value-addition industries to reduce post-harvest losses (**Dr K.S. Varaprasad**).
- Mechanize cotton cultivation and adopt precision farming (**Dr. Y.G. Prasad**).

### Research & Innovation

- Raise agri-research investment to 1.5% of GDP (**Dr. Kareemulla**).
- Strengthen pre- and post-production research (**Dr. Varaprasad**).
- Encourage PPP particularly Industry-driven crop improvement research and agricultural education (**Dr. Gururaj Katti**).
- **National Mission Expansion:** Launch seed treatment as a highly critical national mission to protect crops against seed/soil-borne infections, directly reducing pest and disease losses by 40-50% in the main crop season (**Dr. P.K. Chakrabarthy**).
- **Policy-Technology Integration:** Engage ICAR institutes, SAUs, professional societies, and academia directly in public policymaking to bridge bureaucratic gaps, aiming to elevate climate-enabling policy support from India's current 5% toward the 30% standard of developed nations (**Dr. P.K. Chakrabarthy**).
- **Institutional Training:** Embed mandatory expert lectures covering principles, challenges, and governing policies of plant protection into the Foundation Course for Agricultural Research Service (FOCARS) curriculum at ICAR-NAARM for all incoming interns (**Dr. P.K. Chakrabarthy**).

### Market & Farmer Returns

- Establish mandis near production areas (**Dr. Gopal Lal**).
- Registration system for middlemen to reduce price gaps (**Dr. C.D. Mayee**).

- Support community seed banks for conserving native landraces (**Dr. B. Sarath Babu**).
- Digital awareness campaigns for farmers on plant health and prices (**Dr. Shaik N. Meera**).
- **Academic Defense Platform:** Formulate an organized institutional framework within academic forums to deliver rapid, science-based rebuttals and justifications against false news or anti-science media propaganda that compromises public trust (**Dr. P.K. Chakrabarthy**).

#### **Regulation & Quality Control**

- Monitor sale of unregistered microbial biostimulant products (**Dr. T.V.K. Singh**).
  - Establish National Bioinput Regulatory Authority.
  - Create National Plant Protection Diagnostic Network (**Dr. Parameswari**).
  - **MRL Harmonisation:** Enforce crop-group-wise Maximum Residue Limits (MRLs) and implement crop grouping to completely mitigate off-label pesticide usage, which currently affects a massive 80% of domestic crops against only 15-20% approved use (**Dr. P.K. Chakrabarthy**).
  - **Biopesticide Mainstreaming:** Establish state-of-the-art government Centers of Excellence for rigorous quality assessment and quality certification of biological technologies to resolve decades-long supply constraints (**Dr. P.K. Chakrabarthy**).
  - **Regulatory Acceleration:** Overcome bureaucratic roadblocks and 'fear of the unknown' by immediately approving necessary defoliants required for High-Density Planting Systems (HDPS) mechanical cotton picking, and commercializing HTBt cotton to slash manual weeding costs by 35-40% (**Dr. P.K. Chakrabarthy**).
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# Major Outcomes & Policy Directions (PPAI Institutional Blueprint)

The Plant Protection Association of India, with support from Expert Scientists including policy experts, vice chancellors, KVK and FPOs from all over the country and major industry players, outlined **structured policy outcomes** to be advanced to government agencies:

## 1. Macroeconomic Policy Draft

- A unified macroeconomic roadmap for fiscal and policy adjustments at central and state levels.
- Focus on farmer income protection, diversification incentives, value-chain strengthening, and the integration of Micro-Irrigation as a core policy pillar for national climate resilience and water security.
- **Academic Synthesis:** Institutionalized framework ensuring that ICAR, SAUs, and professional societies actively anchor national agricultural and climate policies to align tech development with active legislation.

## 2. Regulatory Framework

Blueprint for a factory inspection task force to eliminate spurious crop-protection inputs, modelled on pharmaceutical industry standards.

- **Pharma-Grade Auditing:** GMP standards, automated batch logging, unannounced audits.
- **Stage-Gate Traceability:** QR-coded batch fingerprinting, geographic tracking, farmer validation via SMS/app.
- **KYC-Linked Licensing:** Mandatory digital auditing, automated revocation of non-compliant licenses, accountability tied to all stakeholders as per Insecticides Act 1968 & Rule 71.
- **Biotech & Chemical Reforms:** Fast-track clearance protocols for crop-grouped MRLs to slash off-label usage, institutional certification for biopesticides via Centers of Excellence, and time-bound approvals for critical inputs like HTBt traits and cotton defoliants.

### Execution Phases:

- Phase 1 (Months 1-3): Finalize audit checklist, establish central task force.
- Phase 2 (Months 4-6): Pilot digital batch-tracking in major hubs - QR Code is mandatory.
- Phase 3 (Months 7-12): Pan-India rollout, mandatory KYC, which is already rolled out by Ministry of Agriculture & Farmers' Welfare needs proper execution for shutdown of non-compliant units.

## 3. Actionable Research & Training Roadmap

- Coordinated plan for KVK networks and FPO leaders to implement high-yield, sustainable practices in support with Expert Scientists & Private Sector.
  - Emphasis on crop diversification, digital agriculture, and farmer training.
  - **NAARM-FOCARS Integration:** Operationalizing mandatory plant protection policy modules for all incoming Agricultural Research Service interns.
  - **Scientific Truth Broadcasting:** Activating academic media-monitoring cells to counter public misinformation with hard scientific data.
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# Agricultural Transformation Roadmap - Timeline Infographic

## Implementation Framework

A practical implementation framework that maps the integrated outcomes of the brainstorming session into **short-, medium-, and long-term priorities** for policymakers and industry leaders. This structure turns the deliberations and PPAI's blueprint into a roadmap for action.

### Short-Term (0-1 years)

- **Regulatory Enforcement:** Central inspection task force, pharma-grade audits, QR-coded product tracking, KYC-linked licensing.
  - **Farmer Support:** Digital validation tools for inputs, community seed banks, awareness campaigns, and land records.
  - **Water & Irrigation:** Subsidies >50% for micro-irrigation, crop-specific drip packages, groundwater recharge structures (p. 4). Implement smart and digital irrigation tools, and reform subsidy norms to prioritize small, marginal, and vulnerable farmers.
  - **Immediate Approvals & Training:** Clear defoliants for HDPS cotton picking; initiate the NAARM plant protection lecture modules for ARS interns; deploy the academic media-rebuttal platform.
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### Medium-Term (1-3 years)

- **Crop Diversification:** Divert paddy area to pulses/oilseeds, MSP for perishables, value-addition industries, mechanized cotton.
  - **Market Infrastructure:** Mandis near production zones, registration of middlemen, stronger supply/value chains.
  - **Research & Innovation:** Raise agri-research investment to 1.5% GDP, PPP in biotech and various agri-sectors.
  - **National Seed Treatment Mission:** Roll out full-scale seed treatment protocols nationwide to mitigate 40-50% of seasonal pest losses.
  - **Standards & Biotech:** Launch Centers of Excellence for biological technology certification; enforce crop-grouping MRLs to systematically eliminate the 80% off-label pesticide anomaly; clear commercialization of HTBt cotton to reduce manual weeding costs by 35-40%.
  - **MI Expansion:** Scaling MI deployment across water-stressed, drought-prone, and canal-irrigated command areas.
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### Long-Term (3-5 years)

- **Policy & Governance:** Unified macroeconomic roadmap, district-level water authorities, National Bioinput Regulatory Authority. Full institutionalization of MI as a National Water Security & Climate Resilience Strategy.
- **Structural Policymaking:** Integrate ICAR, SAU, and professional society scientists directly into the bureaucratic policy-making matrix to match developed nations' 30% climate policy-enabling standard.

- **Structural Transformation:** Voluntary collectivization of landholdings, youth farmer incentives, empowered KVKs and FPOs.
- **Vision for Viksit Bharat @2047:** Innovation-driven agriculture, counterfeit-free inputs, empowered farmers, global leadership in sustainable farming, and increase export of Agri produce.




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## Policy Briefing Presentation Deck - Enhancing Agriculture's Contribution to GDP

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This outline is ready to be used as a **presentation deck** – each slide is concise, action-oriented, and aligned with government and industry priorities.

### Slide 1 - Context & Objective

- Agriculture contributes **16-18% to GDP**, supporting nearly half of India's population.
- Persistent challenges: soil degradation, irrigation mismanagement, counterfeit inputs, low farmer awareness.
- Objective: **Strengthen agriculture, promote innovation, empower farmers for Viksit Bharat @2047.**

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### Slide 2 - Immediate Priorities (0-1 years)

- **Regulatory Enforcement:** Establish central inspection task force under Ministry of Agriculture; pharma-grade audits, QR-coded product tracking, KYC-linked licensing (p. 6).
  - **Farmer Protection:** Digital validation tools for inputs, strengthen community seed banks, awareness campaigns on plant health & market prices, timely insurance claims, precision agriculture (p. 6).
  - **Water & Irrigation:** Subsidies >50% for micro-irrigation in water-stressed districts, crop-specific drip packages for rice/wheat, mandatory groundwater recharge structures (p. 6).
  - **MI Reforms:** Reform subsidy norms for small/marginal farmers; promote smart and digital precision irrigation technologies (p. 6).
  - **Urgent Regulatory & Academic Actions:** Clear essential defoliants for HDPS cotton picking; launch mandatory plant protection policy lectures at NAARM; operationalize academic fora for science-based rebuttals of fake media news.
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### Slide 3 - Medium-Term Priorities (1-3 years)

- **Crop Diversification:** Divert 10-15% of paddy area to pulses/oilseeds, extend MSP to perishables (tomato, mango), promote value-addition industries to reduce post-harvest losses.
  - **Market Infrastructure:** Establish mandis near production zones, registration system for middlemen, strengthen supply/value chains for perishables.
  - **Research & Innovation:** Raise agri-research investment to 1.5% GDP; PPP in biotech & irrigation technologies; National Plant Protection Diagnostic Network.
  - **National Seed Treatment Mission:** Fully execute as a national mission to minimize seed/soil-borne infections and save 40-50% of pest management costs.
  - **Biotech & Pesticide Overhaul:** Harmonize crop-wise MRLs to end the 80% off-label pesticide usage crisis; establish government Centers of Excellence for biopesticide certification; approve HTBt cotton to reduce manual weeding costs by 35-40%.
  - **MI Expansion:** Scale micro-irrigation into drought-prone, high-value crop zones, and canal-irrigated command areas.
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### Slide 4 - Long-Term Priorities (3-5 years)

- **Policy & Governance:** Unified macroeconomic roadmap for agriculture, district-level water management authorities, National Bioinput and new technologies Regulatory Authority.
  - **Policy Inclusivity:** Permanently integrate ICAR, SAUs, and professional societies into public policy-making circles to elevate climate-enabling technology policies to a 30% global benchmark.
  - **Structural Transformation:** Voluntary collectivization of landholdings, incentives for young farmers (revise ARYA scheme), empower KVKs and FPOs as drivers of sustainable farming.
  - **Vision for Viksit Bharat @2047:** Innovation-driven agriculture, counterfeit-free inputs nationwide, empowered farmers with global competitiveness, crop planning as per market demand.
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### Slide 5 - Execution Roadmap

- **Phase 1 (0-3 months):** Establish task force, finalize audit checklist (KYC covers the Checklist), initiate NAARM FOCARS linkages and media rebuttal frameworks.
  - **Phase 2 (4-6 months):** Digital product batch-tracking, training of farmers, and clearance of HDPS cotton defoliants (p. 7).
  - **Phase 3 (7-12 months):** Pan-India rollout, mandatory KYC, shutdown of non-compliant units.
  - **Phase 4 (1-3 years):** Diversification, market infrastructure, large-scale canal-command MI deployment, research expansion, biopesticide Centers of Excellence, HTBt commercialization, and national seed treatment enforcement.
  - **Phase 5 (3-5 years):** Structural transformation, scientific policy integration, Viksit Bharat vision.
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### Slide 6 - Key Takeaways

- Agriculture must evolve from subsistence to innovation-driven growth.
- Protect farmer incomes through policy, regulation, and market reforms.
- Build resilience via diversification, micro-irrigation efficiency (digital and canal

command integration), and enhanced research investment.

- **Comprehensive Defense:** Protect agricultural science from malicious propaganda while eliminating regulatory 'fear of the unknown' to allow advanced biotechnology to reach fields.
- **Long-term vision:** Empowered farmers, sustainable agriculture, global leadership.
- Total Reforms in present laws after review as to support "Ease of doing Business".

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## Conclusion

The brainstorming session produced a comprehensive, triple-layer outcome:

1. Expert deliberations on irrigation, diversification, research, markets, and regulation.
2. An institutional blueprint by PPAI for government adoption, focusing on policy drafts, pharma-grade regulatory enforcement, and actionable research.
3. A modern framework for regulatory clarity and scientific sovereignty—championed by Dr. P. K. Chakrabarthy—which mandates resolving off-label pesticide usage, certifying bioinputs via Centers of Excellence, shielding public faith through academic rebuttals, and integrating agricultural scientists directly into national policymaking setups.

Together with Dr. V. Praveen Rao's expanded framework for micro-irrigation, these form a comprehensive roadmap for enhancing agriculture's contribution to GDP, protecting farmer incomes, and ensuring environmental sustainability.

**CONVENER'S DECLARATION & RATIFICATION** This is to certify that the policy recommendations, regulatory frameworks, and technological roadmaps compiled herein accurately represent the consensus and resolutions achieved during the PPAI Brainstorming Session held on June 6, 2026.

Ratified on this 2nd day of July, 2026.



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Dr. B. Sarath Babu  
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Dr. B. Parameswari General  
Secretary - PPAI  
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